

Álvaro López Álvarez

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LinkedIn, GitHub, Portfolio

Software Engineer with a strong Computer Science foundation and hands-on experience across backend development, distributed systems, full-stack product engineering, and LLM/agentic AI tooling. Primary languages: Golang and TypeScript, with working proficiency in Kotlin, Java, Python, and SQL. Comfortable owning features end-to-end, from event-driven backend architecture to React frontends, and drawn to deep systems engineering – compilers, interpreters, and network protocols built from first principles. Holds a Master's Degree in Computer Engineering (final grade 8.8/10) and certified B2 English, and actively builds independent projects.

Core Competencies

- **Backend & Languages**: Golang, TypeScript/JavaScript, Java, Python, Kotlin, SQL, GDScript, C
- **Distributed Systems**: Docker, NATS, Apache Kafka, Apache APISIX, event-driven microservices
- **Full-Stack Development**: React, PostgreSQL, REST APIs, OAuth 2.0/JWT
- **LLM/AI Engineering**: LangChain, LangGraph, Gemini API, OpenAI API, RAG pipelines
- **Tools & Practices**: Git, GitHub Actions, Docker Compose, Testcontainers, TDD, Firebase, Supabase, Vercel, Render, Android Studio, NeoVim

Professional Experience

- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  R&D Developer — *Feb 2026–Jun 2026*
  - Implemented LLM-based improvements to a search engine
  - Built a LangGraph multi-step agent that converts natural-language questions into validated, read-only SQL queries to resolve data-access policies over a PostgreSQL data-governance schema
  - Applied safety-conscious agentic design: schema/table selection, structured SQL generation, execution validation, retry-on-error
  - Worked within an application focused on management, governance, and sovereignty of industrial data

- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  Full-Stack Developer Intern — *Feb 2024–Jun 2024*
  - Developed and maintained the corporate personnel management application (Intranet)
  - Implemented new frontend features using React and TypeScript, improving internal usability
  - Contributed to backend development, ensuring smooth integration between services

Selected Independent Projects

- **AIDucate** – *Go, Gin, PostgreSQL, React, TypeScript, Tailwind, Gemini API*: AI-powered SaaS platform generating personalized study plans, content, and exercises. JWT-authenticated Go/Gin backend with PostgreSQL persistence; React/TypeScript frontend deployed on Vercel, backend on Render. *(In progress)*
- **ValenMove** – *Kotlin, MVVM, JSoup*: Native Android app unifying Valencia's public transport (EMT, Metrovalencia, Valenbisi) into one interface; reverse-engineered public APIs and combined web scraping with the city's Open Data portal for real-time arrivals. Final Degree Project, grade 9/10.
- **FaaS with Golang** – *Go, Docker, NATS JetStream, Apache APISIX*: Distributed Function-as-a-Service platform where Go workers execute arbitrary Docker-image functions. NATS JetStream work-queues for reliable task distribution, Apache APISIX gateway for JWT auth and rate limiting, horizontal scaling via Docker Compose replicas.
- **Kafka + Spring Boot Ticketing Platform** – *Java, Spring Boot 3, Apache Kafka, Testcontainers*: Event-driven microservices (reservation, payment, notification, audit) using the outbox pattern, dead-letter topics, idempotent processing, and integration tests with Testcontainers.

- **Monkey Interpreter** – *Go*: Tree-walking interpreter for a custom language, built from the book *Writing an Interpreter in Go*: hand-written lexer, Pratt parser, AST, evaluator, and REPL.

Education

- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Master's Degree in Computer Engineering (MUIINF) *2024–2026*
  Completed (Sep 2026), final grade 8.8/10. Topics: distributed systems, high-performance computing, cybersecurity, embedded systems; coursework includes Embedded and Ubiquitous Systems, Networks and Security.
- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Grado (BSc) in Ingeniería Informática, Software Engineering track *2020–2024*
  GPA: 8.88/10. Thesis (TFG): "ValenMove – Unified Public Transport App" (grade 9/10).

Languages

- Spanish (native), Catalan (native), English (certified B2, confident conversational).

References

- Available upon request.